

KILNDECK · CALCULATION NOTE

KilnDeck sizing and first-principles checks

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	First TRL 3 sizing note on the constructable design (KND-DDR-002), against every requirement

KilnDeck sizing and first-principles checks

On paper the deck is strong, stiff and light enough, easy to move and cheaper than the value-engineering target, but its floor is too hot in full summer sun. Ten of the thirteen requirements in KND-REQ-001 are met on paper. R7 (floor at 50 °C or less) is not met in full sun at 45 °C air, where the floor top reaches about 74 °C, although that is about half the 150 °C of the crust the fireman stands on today. R1 is met on the straight sides only. R8 is met only against an interim bearing limit until the partner kiln is surveyed, and R5 (dose repeatability) needs a weighing trial at TRL 4. The parts cost is USD 2,627. Value-engineering target: USD 3,500. Estimated cost of the constructable design: USD 2,627 (USD 873 under the target).

Every number in this note is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`). The script reads the geometry from `PARAMS` and `levels()` in `cad/src/model.py`, takes masses from the model solids and reads prices from `bom/bom.csv`, so the model, the drawing KND-DWG-001, the BOM and this note agree. All values are first-principles estimates. Nothing here is measured.

■ 1. Assumptions

Table 1. Assumptions. Kiln values are working assumptions until the survey of the partner kiln.

Quantity	Value	Basis
Kiln output	30,000 bricks a day	Midpoint of 20,000 to 50,000 (Greentech, 2014)
Fired brick	3.0 kg	230 x 115 x 75 mm at about 1.5 kg/L
Specific energy	1.2 MJ per kg of fired brick	Assumption for a fixed-chimney kiln
Coal	22 MJ/kg; bulk 0.80 kg/L	Assumption
Biomass	bulk 0.25 kg/L	Chopped stalks or husk, assumption
Feeding	every 17.5 min by two firemen; 18 holes per fireman per round	Greentech (15 to 20 min); 3 rows of 6 holes is an assumption
Trench and track	trench 6,000 mm; track centres 6,600 mm	Assumption until the survey
Feed holes	150 mm across, about 1,000 mm apart	Assumption
Crust surface	150 °C, 250 °C at hot spots	Assumption, to be measured with a thermal camera
Air	45 °C summer day, 30 °C night; sun 1,000 W/m ²	Typical South Asian summer
Live load	fireman 100 kg; rated 250 kg; dynamic factor 1.25	R3
Rolling resistance	2 % (ball-bearing wheels on a dusty bar)	Assumption
Friction of a locked wheel	0.15 (dusty steel on steel)	Assumption
Chain efficiency	85 %	Assumption
Wall bearing	interim limit 0.20 MPa under a pad	Conservative for old brick in mud or lime mortar, until the survey (KND-DDR-002, A2)
Steel	S275, yield 275 MPa, E = 210 GPa	

■ 2. Fuel per round

Coal per day is $30,000 \times 3.0 \text{ kg} \times 1.2 \text{ MJ/kg} / 22 \text{ MJ/kg}$ = about 4,909 kg. With a round every 17.5 min there are about 82 rounds a day, so each round takes about 59.7 kg, or about 29.8 kg per fireman and about 1.66 kg per hole.

■ 3. Hopper (R6)

The hopper holds about 52.1 L below the grille: about 41.7 kg of coal or about 13.0 kg of biomass. One fill covers about 1.4 coal rounds for one fireman. **R6 met** for coal. With biomass the fireman refills every round, which the outer-end refill point allows.

■ 4. Metering rotor (R5)

The four-pocket rotor (160 mm across, 200 mm long, 40 mm hub, 8 mm vanes) moves about 3.08 L per turn, 0.770 L per pocket. At 80 % fill that is about 0.49 kg of coal per pocket (1.97 kg per turn) or about 0.15 kg of biomass. A hole needs about 3.4 pockets of coal a round. The largest lump that passes a pocket is about 40 mm, and the grille gaps are 44 mm. **R5** (repeatability within 15 %) cannot be judged on paper; it is weighed at TRL 4.

■ 5. Masses (R12)

Table 2. Masses from the model solids (kg).

Item	Mass
Deck (trucks, wheels, trusses, bearers, floor, end frames, grips, drive, carriage rail)	about 797
Hopper carriage, hopper, rotor and spout, empty	about 48
Track, pads and stops for two 12 m runs	about 358
Heaviest pieces carried up	truss module B 39.2; truss module A 38.8; outer end truck 37.8; track module 33.4; floor panel 30.4; frame bearer 6.9

R12 met: no piece is over 40 kg. The truss modules have under 1 kg of margin, so heavier tube than specified must not be substituted.

■ 6. Bridge (R3)

Each side truss is checked as a simply supported beam over the 6,600 mm track centres, with chords 1,143 mm apart (second moment about 29,024 cm⁴ per truss). Truss B takes half the deck's own weight (0.537 N/mm), the carriage with a full hopper (about 90 kg) and half the fireman, all at mid-span with a 1.25 dynamic factor: 1,717 N.

Table 3. Bridge results.

Case	Bending moment	Chord force	Chord stress	Deflection
Truss A in use	3.93 kN m	3.44 kN	8 MPa	0.4 mm
Truss B in use	5.76 kN m	5.04 kN	11 MPa	0.5 mm (span/13,143)
Truss B, proof 1.5 x 250 kg	8.99 kN m		18 MPa	

Deflections include a 1.3 factor for the shear deformation of the truss. The top chord could buckle between U-frames (1,725 mm) at 71.0 kN against 5.04 kN, a ratio of 0.07. Each M12 splice bolt near the bottom chord carries about 2.52 kN in tension. **R3 met** with a large margin; the trusses are sized by the guardrail load and the 40 kg piece limit, not by the bridge.

■ 7. Guardrail (R2)

A 1 kN horizontal push on the top rail is checked in three places. Between posts (clear span 1,645 mm, simply supported) the top chord sees 81 MPa. At a splice, the two bolted end verticals act together as a cantilever 1,143 mm tall: 91 MPa and 9.4 mm of movement at the rail. At the deck ends, the truss end vertical and the end rail or

gate post bend together: 100 MPa. The 1.1 kN m at the foot of a splice post passes into the frame bearer through its bolted end plate, about 7.1 kN in each of the two tension bolts (M12 8.8 holds about 48 kN). At the 1.5 kN proof load the stresses are 136 and 150 MPa, below the 275 MPa yield. **R2 met.**

■ 8. Floor panel

A 1.5 kN heel load in the middle of a panel spanning 858 mm between bearers puts 98 MPa in the two 40 x 40 x 4 edge angles and deflects them 1.0 mm. Under a 60 mm square heel spread by the 2 mm top sheet, the board sees about 0.41 MPa; the board's compressive strength is to be confirmed with the supplier (KND-DEC-001).

■ 9. Wheels, track and walls (R8, R13)

The worst wheel load is about 4,387 N (447 kg), on the side B wheel of the truck nearest the carriage when it is at its travel limit, with the dynamic factor. The wheels are specified at 500 kg or more. The track channel between pads 1.0 m apart (UPN 80 bending about its weak axis, bar ignored) sees 172 MPa and deflects 2.24 mm under that wheel. The pad below carries about 4,496 N, a bearing pressure of about 0.150 MPa against the interim limit of 0.20 MPa. **R8 met against the interim limit only;** it is confirmed when the survey sets the real limit.

Across the kiln the deck cannot tip: the restoring moment of its own weight about the side B wheel line is 4.69 kN m against 0.36 kN m from the carriage. **R13** is met by the end stops, buffers and derailment guards (KND-DDR-002, C2 and C3).

■ 10. Travel and parking (R4, R10)

The loaded deck (deck, carriage, full hopper and fireman) weighs about 9,687 N. At 2 % rolling resistance it takes about 194 N to move, which the 45 mm tread radius and the 160 mm hand wheel radius, through 1:1 chains at 85 %, turn into about 64 N at the rim on level track and about 128 N on a 2 % grade. **R4 met.** One turn moves the deck about 283 mm, and the 12 lock holes in the rim give a step of about 24 mm. The 25 mm line shaft sees 2.8 MPa in torsion. The carriage takes about 27 N to push along its rail.

On a 5 % grade the deck pulls 484 N downhill. With the lock pin in, the two driven wheels are locked and carry about half the weight; sliding them takes about 727 N at a friction of 0.15. **R10 met** with a factor of about 1.5.

■ 11. Floor temperature (R7)

A steady heat balance through one square metre of floor: the crust radiates to the aluminium underside (emissivity 0.3 when dusty, against 0.9 for the crust) and hot air under the deck adds convection at 5 W/m²K; heat crosses 37 mm of calcium silicate board (0.065 W/mK); the top sheet absorbs half the sun and loses heat to the air at 10 W/m²K and by radiation to the sky.

Table 4. Floor top temperature.

Case	Floor top	Underside	Heat through	Bare crust
Night, nominal crust, 30 °C air	34 °C	104 °C	123 W/m ²	150 °C
Summer day, shaded, 45 °C air	48 °C	110 °C	109 W/m ²	150 °C
Summer day, full sun, 45 °C air	74 °C	114 °C	71 W/m ²	150 °C
Hot spot, full sun	81 °C	195 °C	201 W/m ²	250 °C
30 °C day, full sun	62 °C	109 °C	83 W/m ²	150 °C

R7 not met in full sun. The sun, not the kiln, sets the floor temperature by day: in shade the floor holds 48 °C. The design keeps no canopy on the first prototype (KND-DDR-002, A3), and the fireman keeps heat-resistant footwear. The edges of each panel, where the steel frame joins top and bottom, will run hotter than these figures.

■ 12. Spout alignment (R9)

Along the kiln the deck stops within half a lock step (12 mm) plus about 5 mm of chain play: about 17 mm. Across the kiln the carriage is pushed by eye to the hole and clamped: about 10 mm. A 100 mm spout over a 150 mm hole leaves 25 mm each side. **R9 met** on paper.

■ 13. Heat movement

A 3 m track module grows about 3.6 mm when it warms by 100 K; the joints have 6 mm gaps and slotted fishplates. The deck grows about 4.8 mm across the trench when it warms by 60 K; the inner wheels float 12 mm either way.

■ 14. Cost (R11)

The 17 lines of `bom/bom.csv` total USD 2,626.97 (steel at USD 1.30/kg from the model masses; bought parts at regional retail estimates; fabrication labour not included). Value-engineering target: USD 3,500. Estimated cost of the constructable design: USD 2,627 (USD 873 under the target). **R11 met** against the target.

Safety: These are first-principles estimates. The guardrail, deck and track must be proof-loaded (CalRig) and the kiln wall surveyed by a competent person before anyone stands on the deck over a fire. Nothing in this note authorises use.